LINYAN(LYDIA) LI

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EDUCATION

University of Chicago — Chicago, IL
M.S. in Data Science, GPA: 4.0/4.0 — Sep. 2026 – Dec. 2027

University of Wisconsin–Madison — Madison, WI
B.S. in Statistics & Data Science (Double Major), GPA: 3.8/4.0 — Sep. 2024 – May 2026

- **Awards**: Dean's List; Undergraduate Scholarship

RESEARCH EXPERIENCE

Harvard AI and Robotics Lab — Sep. 2026 – Present
Research Assistant | Harvard University | Advisor: Prof. Mengyu Wang

- **Agentic Image Geolocation Privacy:** Developing a multi-agent black-box optimization framework for adversarial perturbation generation against closed-source VLMs, integrating adaptive budget allocation and CLIP/DINO-based perceptual constraints
- **Decision-Making Data Curation Platform:** Developing a web-based data curation platform with standardized data collection protocols, server-hosted infrastructure, and scalable data processing workflows

Computational Optimization Research Program — Mar. 2025 – Nov. 2025
Research Assistant | Advisor: Dr. Yuqiang Ning

- Led a lane-level traffic assignment study using MSA and SUMO for congestion and equilibrium modeling
- Co-authored *From Links to Lanes*, accepted for the 2025 INFORMS Annual Meeting

Healthcare Operations Management Excellence LAB — Nov. 2023 – Jan. 2025
Research Assistant | University of Nottingham | Advisor: Prof. Lei Liu

- Focused on computational optimization and presented at the 4th NUBS Tri-Campus Conference as the only undergraduate presenter

WORK EXPERIENCE

YG Technology Co., Ltd. (Incubated by the Tsinghua University Laboratory of Brain & AI) — Beijing, China
Data Science Research Intern — *Jul. 2026 – Aug. 2026*

- Built an **AWS S3 ETL pipeline** integrating multimodal EEG, heart rate, eye-tracking, and behavioral data from **900 sessions** across **39 participants**, generating **4,366** analysis windows
- Developed **multimodal learning pipelines** using Random Forest, early/late fusion, and meta-learning, achieving an **F1 score of 0.93** with participant-level GroupKFold validation

Zhongguang Suchuang Technology Co., Ltd. — Chongqing, China
Data Science Intern — *Jun. 2025 – Aug. 2025*

- Cleaned and aggregated **50K+ transaction records** with window functions and CTEs; engineered weekly demand, lag, rolling, and seasonality features and compared ARIMA/Prophet forecasts using rolling-window validation to avoid look-ahead bias
- Designed a four-metric evaluation framework pairing inventory turnover as the primary objective with stockout rate and revenue stability as guardrails, preventing turnover gains that degraded service levels

Kantar Worldpanel — Guangzhou, China
Data Analytics & Quantitative Research Intern — *Jun. 2023 – Aug. 2023*

- Built Python/SQL ETL and feature pipelines for **2,000+ e-commerce livestreams**, combining engagement, traffic, conversion, and purchase-history signals; trained and tuned XGBoost/LightGBM classifiers, achieving **0.86 F1** and a **12% recall lift** over the baseline
- Converted model scores into ranked high-value customer segments for P&G targeting and analyzed performance by SKU, channel, and purchase frequency to support quarterly category and campaign decisions

Boston Consulting Group — Shanghai, China
Management Consulting Intern — *Jan. 2023 – Mar. 2023*

- Integrated multi-source automotive aftermarket data with Python/SQL; applied log-log regression and K-modes clustering across **10+ product and customer segments** to estimate price elasticity and identify channel-specific purchasing patterns
- Built market-sizing, unit-economics, and sensitivity models using three years of industry data, translating the results into opportunity sizing and capability-gap recommendations for a client's NEV strategy

PUBLICATIONS & CONFERENCE

- **Linyan Li, Yuqiang Ning (Co-Author)**, "From Links to Lanes: A Lane-Level Traffic Assignment Framework." *Accepted for presentation at the 2025 INFORMS Annual Meeting, Atlanta, Georgia, USA, Oct 2025*
- **Linyan Li, Lei Liu (Advisor)**, "A Reinforcement Learning Approach to Dynamic Pricing of Perishable Products." *Presentation at the 4th Nottingham University Business School Tri-Campus Conference, Nottingham, UK, April 2024.*

PROJECTS

Interpretable Multimodal Meme Learning | *PyTorch, CLIP, SAE, Transformer* Mar. 2026 – May 2026

- Developed a **CLIP-based multimodal representation learning pipeline** on **20,827 memes**, using a Top-K Sparse Autoencoder to identify **2,493 concept-aligned features** (**6.5**× improvement in interpretability)
- Applied **Fisher's exact test** with BH correction and conducted **feature ablation and activation steering** to validate interpretable concepts and causal effects across hate, humor, and stance prediction tasks

Network Optimization for Charging Station Placement | *Julia, JuMP, HiGHS, MILP* Feb. 2026 – May 2026

- **Optimization Modeling:** Formulated portable charging-station placement as five mixed-integer optimization models integrating shortest-path routing, set covering, battery dynamics, budget constraints, and equity objectives.
- **Network Formulation:** Built a graph-based campus network with 20 nodes, 32 pedestrian edges, and 36 OD pairs, integrating facility location, network flow, and battery constraints into a unified optimization framework.
- **Solution & Validation:** Implemented the framework in Julia/JuMP with the HiGHS solver and conducted sensitivity and robustness analyses, identifying a five-station deployment achieving 100% route coverage

Large-Scale Parallel Review Modeling | *Python, Linux, XGBoost, HTCondor, CHTC* Feb. 2026 – Apr. 2026

- **Large-Scale Data Engineering:** Built a distributed pipeline on the CHTC cluster using HTCondor to process 100M+ Steam reviews, partitioning the dataset into approximately 100 CSV chunks for parallel execution and improving end-to-end preprocessing and feature engineering efficiency by approximately 20×.
- **Helpfulness Prediction Modeling:** Developed XGBoost and regression models with 40 engineered features, achieving RMSE = 0.86 and R^2 = 0.71 on a 20M-sample set and a 22% RMSE reduction over a regression baseline.

Lane-Level Traffic Assignment Framework | *Python, SUMO, Network Optimization* Jun. 2025 – Oct. 2025

- **Lane-Level Modeling:** Developed a lane-level dynamic traffic assignment framework by representing traffic networks as directed lane graphs and explicitly modeling straight, turning, and lane-changing maneuvers.
- **Congestion Modeling:** Proposed maneuver volume–delay functions incorporating origin- and receiving-lane congestion, modeling spillback, merging friction, and lane-changing interactions ignored by link-based models.
- **Equilibrium Optimization:** Applied MSA and lane-level shortest paths, validated with SUMO simulations and the Sioux Falls benchmark network, improving accuracy by 22%.

Agent-Based Scheduling Optimization | *Python, CPLEX, Zhipu API, Chain-of-Experts* Sep. 2024 – Jan. 2025

- **Intelligent Scheduling Framework:** Developed a modular end-to-end scheduling framework using Python and LangChain, leveraging GLM-4 to parse surgical cases and integrate LLM reasoning with operations research optimization for automated case analysis, surgical scheduling, and medical resource allocation. Prompt compression reduced token consumption and inference latency for real-time deployment.
- **Multi-Agent Optimization:** Designed a Chain-of-Experts framework translating hospital policies and physician constraints into MILP templates solved by CPLEX. Evaluated on scenarios with 20 operating rooms and 1,000+ surgical cases, improving scheduling efficiency by 18%, resource utilization by 13%, and decision speed by 20%.

RL for Dynamic Pricing | *Python, Deep Q-Learning, Actor–Critic, LSTM* Nov. 2023 – Apr. 2024

- **Sequential Decision Modeling:** Formulated dynamic pricing and inventory control as a Markov Decision Process, modeling inventory, selling horizon, demand, price elasticity, and forecasts as states and pricing decisions as actions while designing a risk-adjusted reward to balance profit maximization and inventory risk.
- **Policy Training and Backtesting:** Trained and evaluated Deep Q-Learning, Actor–Critic, and LSTM-based policies, improving gross profit margin by approximately 12% while reducing inventory loss by approximately 15% in backtesting experiments.

SKILLS

Programming & Data Engineering: Python, SQL, R, Julia, Linux, Git, Pandas, NumPy, SciPy
Machine Learning & Statistics: PyTorch, Scikit-learn, XGBoost, Logistic Regression, Random Forest, Time Series, PCA, GMM
Generative AI & Optimization: Transformer, CLIP, Sparse Autoencoders, LangChain, LLM Agents, CPLEX, JuMP, HiGHS, MILP
Analytics & Visualization: Tableau, Plotly, Matplotlib, Experimental Design, Model Validation